Question

Create a file with at least two examples explaining all operators, loops and functions

Answers

OPERATORS

Definition: these are symbols that tell a computer to perform a calculation e.g checking if one number is greater than another

Examples

Arithmetic operators:

- Addition (+): this adds two variables
- Floor division (//): This divides two values but only gives you the answer and not the remainder

Assignment operators: these are symbols used to store values in variables

- Assignment operator (=): this assigns a value to the named variable e.g. a=3
- Addition assignment (+=): this adds a value to the previous value and the sum becomes the new value

Python comparison operators: these compare two values or variables in programming

- Equal to (==): this returns a true is both values are exactly the same e.g. a=4,b=4, then a=b is a true statement
- Not equal to (!=): returns true if the values are not the same

Membership operators: these are used to check whether a value or variable is present in a data collection

- (in): if the variable you're looking for is present in the sequence then it give a true statement

Bitwise operators: symbols used to manipulate the individual binary digits of integer values

- Bitwise and (&): compares corresponding bits and results in 1 only if both bits are 1

Loops

Definition: a loop is an instruction given to the computer to repeat an action until a certain condition is met

For loop: this executes a block of code repeatedly once for each item in the collection

It only runs code for a certain number of times

While loop: it runs a block of code until a certain condition is met

e.g.

```
counter = 0

while counter < 3:

        print ("inside loop")

        counter = counter + 1

else:

      print ("inside else")
```

Functions

Definition: this is a named group of statement that performs a specific task

A function is created by a keyword called a def followed by a valid name of a function (variable syntax)

A function is self contained in a way that it cannot be accessed

```
def staff ():
   num1, num2 = 10, 20
   print (num1 + num2)
   num1, num2 = 100, 200
   print (num1 + num2)
staff ()
```

A return key/statement word makes a function to share a value outside of it

it also marks the end of execution of a function

```
   print (f"your pay before tax is:{pay}")
grosspay (400000)
print (grosspay (50000))
def tax(rate):
   return rate
def cal():
   tax_amount = grosspay(500000*tax (0.3))
   return tax_amount
```

```
def netpay():
    print(grosspay(500000)-cal())
netpay()
```